//SIT55 - OOP ASSIGN 2

/*PROGRAM STATEMENT : Identify commonalities and differences between Publication, Book and Magazine classes.
Title, Price, Copies are common instance variables and saleCopy is common method. The differences are, Bookclass has author and orderCopies(). Magazine Class has methods orderQty, Current issue, receiveissue().Write a program to find how many copies of the given books are ordered and display total sale of publication*/

/*POLYMORPHISM : Polymorphism in Java is the task that performs a single action in different ways.
Two types of polymorphism are : Runtime and Compile time. Runtime polymorphism is achieved through method overriding,
and compile-time polymorphism is achieved through method overloading*/

```
package Javaprogs;
class Publication
{
   public String title; //data members
   public double price;
   public int copies;
   public Publication(String t,double p,int c) //parameterized constructor is used to initialize objects
   {
      this.title=t; //'this' keyword is reference variable that refers to the current object
      this.price=p;
      this.copies=c;
   }
   public void sellcopy() //method of class
   {
      System.out.println("TOTAL PUBLICATION SALE = "+(copies*price));
   }
}
class Book extends Publication /*inheritance is achieved by using keyword "extends".
It indicates that class Book is derived from baseclass Publication.
"extends" keyword Extends the functionality of the parent class to the subclass.*/
{
   private String author;
   public Book(String t,double p,int c,String a) //parameterized constructor is used to initialize objects
   {
      super(t,p,c); /*super keyword is used to call superclass i.e.parentclass methods, and
      to access the superclass constructor.It is used to avoid confusion between superclasses and
subclasses that
      have methods with the same name*/
      this.author=a;
   }
   public void ordercopies(int b) //method of class
   {
      System.out.println("ORDER OF BOOK = "+b);
      this.copies=b;
```

```
    }
    public void sellcopy() /*method overriding is achieved.
    If subclass has the same method as declared in the parent class, it is known as method overriding.
    It provides specific implementation of method that is already provided by its superclass*/
    {
        System.out.println("TOTAL BOOK SALE = "+(copies*price));
    }
}
class Magazine extends Publication  //inheritance is achieved by using keyword "extends".
{
    private int orderQty;
    private String currentissue;
    public Magazine(String t,double p,int c,int o)  //parameterized constructor is used to initialize objects
    {
        super(t,p,c); //super keyword is used to call superclass methods and to access the superclass
constructor
        this.orderQty=o;
    }
    public void receiveissue(String c) //method of class
    {
        this.currentissue=c;
        System.out.println("CURRENT ISSUE = "+currentissue);
    }
    public void sellcopy() //method overriding is achieved.
    {
        System.out.println("TOTAL MAGAZINE SALE = "+(orderQty*price));
    }
}
public class Polymorphism
{
    public static void main(String args[])
    {
        Publication p = new Publication("Lifespace", 133.22, 66); /*object of class Publication is created
and
        as soon as object is created ,constructor of that class is called*/
        Book b = new Book("Atomic Habits", 200, 50, "Robin"); //object created
        Magazine m = new Magazine("Godhead", 62, 20, 45); //object created
        p.sellcopy(); //methods are called by objects of class
        b.ordercopies(50);
        b.sellcopy();
        m.sellcopy();
        m.receiveissue("ABC 6.6");}
}

/*OUTPUT :
TOTAL PUBLICATION SALE = 8792.52
ORDER OF BOOK = 50
TOTAL BOOK SALE = 10000.0
TOTAL MAGAZINE SALE = 2790.0
CURRENT ISSUE = ABC 6.6 */
```